F-tests

Ithuteng Ntakhwana ID: 201801847

```
#Load Air passengers dataset embedded in R
passengers = data.frame(year = rep(1949:1960, each = 12),
                        month = c(1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12),
                        fq = c(112, 118, 132, 129, 121, 135, 148, 148, 136, 119, 104, 118,
                                 115, 126, 141, 135, 125, 149, 170, 170, 158, 133, 114, 140,
                                 145, 150, 178, 163, 172, 178, 199, 199, 184, 162, 146, 166,
                                 171, 180, 193, 181, 183, 218, 230, 242, 209, 191, 172, 194,
                                 196, 196, 236, 235, 229, 243, 264, 272, 237, 211, 180, 201,
                                 204, 188, 235, 227, 234, 264, 302, 293, 259, 229, 203, 229,
                                 242, 233, 267, 269, 270, 315, 364, 347, 312, 274, 237, 278,
                                 284, 277, 317, 313, 318, 374, 413, 405, 355, 306, 271, 306,
                                 315, 301, 356, 348, 355, 422, 465, 467, 404, 347, 305, 336,
                                 340, 318, 362, 348, 363, 435, 491, 505, 404, 359, 310, 337,
                                 360, 342, 406, 396, 420, 472, 548, 559, 463, 407, 362, 405,
                                 417, 391, 419, 461, 472, 535, 622, 606, 508, 461, 390, 432
                        ))
write.csv(passengers,"Airlinepassengers.csv", row.names = FALSE)

#first rows of passengers dataset
head(passengers)
```

```
##   year month  fq
## 1 1949     1 112
## 2 1949     2 118
## 3 1949     3 132
## 4 1949     4 129
## 5 1949     5 121
## 6 1949     6 135
```

```
#stucture of dataset
str(passengers)
```

```
## 'data.frame':    144 obs. of  3 variables:
##  $ year : int  1949 1949 1949 1949 1949 1949 1949 1949 1949 1949 ...
##  $ month: int  1 2 3 4 5 6 7 8 9 10 ...
##  $ fq   : num  112 118 132 129 121 135 148 148 136 119 ...
```

```
#summary of dataset
summary(passengers)
```

```
##       year          month             fq
##  Min.   :1949   Min.   : 1.00   Min.   :104.0
```

```
##  1st Qu.:1952   1st Qu.: 3.75   1st Qu.:180.0
##  Median :1954   Median : 6.50   Median :265.5
##  Mean   :1954   Mean   : 6.50   Mean   :280.3
##  3rd Qu.:1957   3rd Qu.: 9.25   3rd Qu.:360.5
##  Max.   :1960   Max.   :12.00   Max.   :622.0
```

```
#check for missing values
library(zoo)
```

```
## Warning: package 'zoo' was built under R version 4.2.3
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
passengers = na.approx(passengers)
```

```
#simulated data for comparison
simulations = rnorm(length(passengers), mean(passengers), sd(passengers))
simulations
```

```
##   [1]  -132.406612  1028.777781   662.082873  1929.199513  1313.184224
##   [6]  1303.347927  -515.765054  -368.903511   147.579859  1680.646254
##  [11]  -516.665854  -168.597191  -113.199134   450.072515 -1591.882598
##  [16]  -142.346365  1576.239559  1149.932154   413.443473   287.222288
##  [21]  1662.701473   385.782596  1464.662786   331.660128  2294.510288
##  [26]   705.127398  -301.197278  1462.039601  1064.211858  1112.757172
##  [31]  -541.868376   927.692771  -865.109552  -106.435953   272.949785
##  [36]   394.388663   779.593944  1119.155317    28.127736  -399.594080
##  [41]  1968.961153  1852.620073 -1000.740177  1663.603311  1658.103801
##  [46]  1634.932105   652.642609   292.322021   171.899192   913.708748
##  [51]   855.586435   355.197856  1413.924266  1601.061713   558.707572
##  [56]   456.211087  -211.956810   472.122381   597.510828   215.926197
##  [61]   219.045520   500.602747 -1079.082949  2036.209333   954.019883
##  [66]  -437.372116  1939.072315  1253.758061  -248.846672  1249.790966
##  [71]    31.559458    68.646464  1395.850867   692.387483  1582.993813
##  [76]  -128.817021   222.171112  1751.079786   484.383397  -500.162564
##  [81]  1441.667513  -165.602751  -286.416673  2198.186970  1041.188769
##  [86]  2080.430952   800.209689   550.382732  -426.912841  1917.960353
##  [91]  1534.823299    53.794113  -197.168159  -127.813023  2508.219208
##  [96]  1379.609766  1496.929287   344.080396  1953.713545   885.846533
## [101]   280.836614  1493.154217  3131.666832  -481.979069  1627.437409
## [106]  1596.915578  1295.521202   655.137049  -133.513109  -480.639342
## [111]   527.423307   595.377928   485.081524   856.436552  -508.314265
## [116]    34.765909  -342.581619  -496.141818   757.680648  1062.214393
## [121]   403.406219   -26.133088  1115.823609   666.304572  1433.403796
## [126]  1355.230869   890.040464     6.400238   675.475324  1441.945784
## [131]  -169.264990  -184.506698  1778.805257  -343.812916  2195.826541
## [136]  1162.093248   939.762601  -146.241493  1971.153023  1124.016455
```

```
## [141]  -269.974308   574.941035   313.894724   -93.109643  1076.139956
## [146]  1543.370738   203.594738  -873.420919   883.003464  1677.611256
## [151]  1467.697682   992.180652   973.472710  1265.846862  1373.599896
## [156]  -153.460727 -1295.288742  1193.524217  -403.571944  2137.972876
## [161]   772.090423   737.834874  2182.772958  1036.752640   741.021320
## [166]   636.179612   356.284048   273.142486  -365.126080   847.686840
## [171]  2011.537760   356.877436  1923.693280 -1231.943526   -13.516195
## [176]   152.161984   395.396072   502.216932   261.227382  2489.216552
## [181]   134.062170  1217.806031  1668.181821   498.505365   779.826603
## [186]  3168.968803  1814.895531  1838.923594  1008.345424  1354.995505
## [191]   395.671340   938.547810   -36.092362  -237.894934  -828.412605
## [196]    68.831380   138.392609   525.767262   361.483942  1724.253422
## [201]  -401.940616   948.976630   463.375703   371.072009   180.156983
## [206]   -18.946785  -511.770147  1334.451355  1181.254976   771.257454
## [211]  -666.683393  1908.113151  1209.922264  2088.829442  1041.681872
## [216]  -128.021536  1093.135014   518.536410  1611.298078  2923.302708
## [221]  1582.040278   815.701716  -287.214065  1099.871170  2899.991409
## [226]   621.051509  1125.880167  -238.529763  -856.836918   517.779799
## [231]  1516.685445  1348.376214   -14.279061   235.782753   610.164880
## [236]   -23.253067  1769.954569  -290.041648   420.566255    56.582319
## [241]   -39.986353  1054.076751     9.081687  2638.710144   786.653974
## [246]   512.621228  1601.653067  2006.599408  1585.308256   105.954229
## [251]   646.015844  1958.604017   598.523821  1464.403921 -1617.372025
## [256]  1512.876379    22.788493  1501.868216  -122.909557    58.559191
## [261]  2374.088785   817.486350  1593.060126   355.536775   920.561323
## [266]   692.655243   326.299963   127.182010  1511.004884   663.229384
## [271]   691.779715  -270.621891  1575.317646  1583.344495 -2688.173873
## [276]  1227.381484  1389.107711   599.050419   965.308756  2073.296100
## [281]  1072.391344  2076.693630   147.167067    90.518149   218.324375
## [286]   499.647701  1189.037377  1316.848580   742.705534  1429.323708
## [291]    -3.071539  1281.464367  -184.970878   316.816151  -570.881643
## [296]  1121.750083   834.838256  1504.694410  2149.060474   758.305497
## [301]  1386.194033  3403.875688   -57.537643  1975.481800  -589.302040
## [306]   328.972503   364.040990   782.134086  1284.649430    89.333213
## [311]  1931.454441   495.382266   733.004083   789.622470   638.829520
## [316]   408.172097   587.047966  -756.254299   210.544225  1835.443376
## [321]   179.060846   501.083279   210.380013   176.829822  -304.986290
## [326]   589.144212   615.694585  1512.679866   144.903734  1618.350776
## [331]  -494.992013  -352.868968  1136.692611  2097.275104   653.601830
## [336]  -138.943779   550.059976  -790.651021   531.617047   509.754196
## [341]   533.603136  2303.325921   833.587346  1381.084652  -707.763971
## [346]  -204.896185   737.107722   504.005137   895.397754   833.979113
## [351]  1975.680980  2250.310197   866.346500  1447.734964   433.723513
## [356]   103.300122  1700.873954   359.168393   258.182092  2056.394799
## [361]  1351.377614  1542.220826  1149.677430  1048.241953  -954.183552
## [366]   218.152798   490.867255  1835.387625  -102.397552  1543.161739
## [371]   629.027093  -330.015628  1000.536446   799.832749    22.336762
## [376]  1010.553019  1992.030628   959.935292  1410.589049  1086.863464
## [381]   326.879335  -298.938667 -1892.306461   138.450826  -282.884700
## [386]   -59.233457  1003.646979   800.188377  1468.715247  1316.898565
## [391]  1617.843695  -112.992761  1743.311259   523.113670  1250.649795
## [396]  1695.658840   359.267460   227.335807  1226.749009   -73.691446
## [401]   347.908563   896.011582   950.425792   809.782494   522.311633
## [406]  2783.926102  1011.738745  1434.469413  1705.672298  1089.450239
```

```
## [411]  1129.064365   477.272145   802.037387  -155.756123   437.066697
## [416]  -267.640251  1220.500502   466.949997 -1170.711461  1691.455421
## [421]   508.700403   388.269501   606.102536 -1563.999667   284.792594
## [426]  2103.445279   312.181538  1609.076643  1436.522626  1884.425235
## [431]  1105.616663 -1117.684849
```

```
#performing an F-test
f_test = var.test(passengers, simulations)
f_test
```

```
##
##  F test to compare two variances
##
## data:  passengers and simulations
## F = 0.97163, num df = 431, denom df = 431, p-value = 0.7653
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.8042517 1.1738507
## sample estimates:
## ratio of variances
##          0.9716334
```

```
#p_value from F-test
p_value = f_test$p.value
print(p_value)
```

```
## [1] 0.7652971
```